

WASP-43b

R-0601

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230215 · created 2026-07-25T03:46:58+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459990.7775 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.057
Transit depth (Rp/Rs)^2	3.38 +/- 2.1 [%]
Orbital Inclination (inc)	77.64 +/- 2.96
Airmass coefficient 1 (a1)	0.61 +/- 0.3
Airmass coefficient 2 (a2)	0.26 +/- 0.42
Scatter in the residuals of the lightcurve fit is	44.28 %
Best Comparison Star	#1 - [589, 162]
Optimal Aperture	1.93
Optimal Annulus	5.75
Transit Duration (day)	0.026 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.057 vs 0.1594 (deviation 0.43σ)
Duration fit vs published	0.026 d vs 0.0512 d (deviation 1.26σ)
Mid-time O–C	5.1 min
Depth SNR	3.2
Residual scatter	44.28 %

Light curve

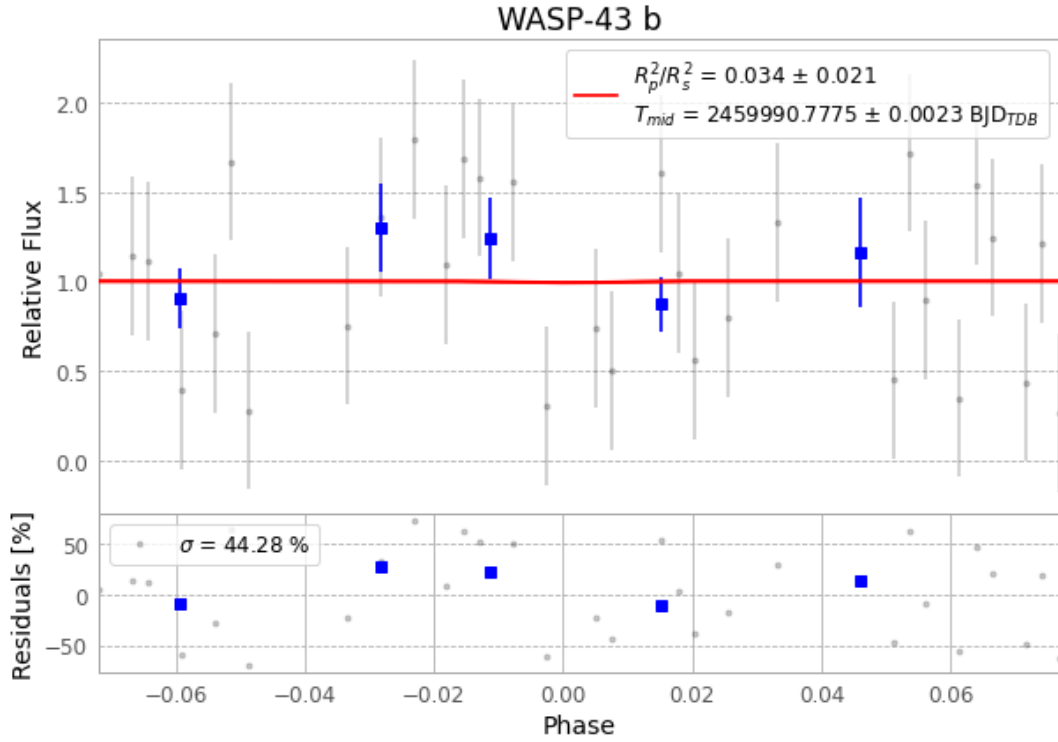


Figure 1 — FinalLightCurve_WASP-43 b_2023-02-14.png (SHA-256 bee619564824adb...

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [109, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 413f75d41c2a1aca...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

63 FITS frames (41 MB), WASP-43230215045415.FITS → WASP-43230215080015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230215.log (baba1b042347...), FinalLightCurve_WASP-43 b_2023-02-14.png (bee619564824...), FinalLightCurve_WASP-43 b_2023-02-14.csv (5a5f6c1e0eb6...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 03:46Z.